

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper: BSC07A / BP Anc.-II: Ancillary Physics-II

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **six** of the following: [6 × 3 = 18]

- (a) What do you mean by intensive and extensive properties?
- (b) Explain the dual nature of light with examples.
- (c) How can diffusion and Brownian motion be explained on the basis of random walk?
- (d) What will be the value of entropy at $T = 0\text{K}$? Will a Carnot cycle work at this temperature?
- (e) Green light of wavelength $5100\text{ \AA}$ from a narrow slit is incident on a double slit. If the overall separation of 10 fringes on a screen 200 cm away is 2 cm, find the slit separation.
- (f) Explain various uses of polarization of light.
- (g) Define the power of a lens. Mention the formula for the power of a combination of two thin lenses.
- (h) What do you mean by thermal equilibrium? Which law of thermodynamics defines it?

Question 2.

- (a) Explain various types of defects in the human eye. How can these defects be removed using different lenses? [5]
- (b) State and explain Fermat's principle of extremum path with an example. [3]
- (c) Explain the differences between the Fresnel and Fraunhofer classes of diffraction. [5]

Question 3.

- (a) Define the resolving power of an optical instrument. Obtain an expression for the resolving power of a telescope. [7]
- (b) A point source of light is located 20 cm in front of a convex mirror with a focal length of 15 cm. Determine the character and position of the image point. [6]

Question 4.

- (a) Calculate the focal length of a plano-convex lens of which the radius of curvature of the curved surface is 40 cm and the refractive index is $\mu = 1.5$. [5]
- (b) Explain the construction and working of a compound microscope using a proper ray diagram. [8]
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Question 5.

- (a) Establish the relations for thermodynamic variables S , T , P , and V in terms of thermodynamic potentials U , F , H , and G , where the symbols have their usual meanings. [8]
- (b) What do you mean by an isothermal process? Show that in an isothermal expansion of an ideal gas, $\Delta U = 0$ and $\Delta H = 0$. [5]
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Question 6.

- (a) Discuss the working of a Carnot engine and sketch its cycle on a P - V diagram. Explain the four processes involved. [6]
- (b) The efficiency of a Carnot engine can be increased in two ways: (i) by decreasing the temperature of the sink T_2 while keeping the temperature of the source T_1 constant, or (ii) by increasing the temperature of the source T_1 while keeping the temperature of the sink T_2 constant. Which of these methods is more effective and why? [7]
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Question 7.

- (a) What is the difference between effusion and diffusion? [3]
- (b) State Graham's law of diffusion. [3]
- (c) What do you mean by drift velocity and mean free path of a gas molecule? [4]
- (d) Why are water droplets formed when a bicycle tube bursts? Explain. [3]
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